

Boundary-constrained chunking shapes local-global positional representations in working memory

Xingyu Yang, Yao Wang & Likan Zhan

To cite this article: Xingyu Yang, Yao Wang & Likan Zhan (2026) Boundary-constrained chunking shapes local-global positional representations in working memory, *Memory*, 34:8, 974-985, DOI: [10.1080/09658211.2026.2712599](https://doi.org/10.1080/09658211.2026.2712599)

To link to this article: <https://doi.org/10.1080/09658211.2026.2712599>

 View supplementary material 

 Published online: 04 Aug 2026.

 Submit your article to this journal 

 Article views: 45

 View related articles 

 View Crossmark data 



Boundary-constrained chunking shapes local–global positional representations in working memory

Xingyu Yang^{a,b,c,d}, Yao Wang^{a,b,c} and Likan Zhan^{a,b,c}

^aCognitive Science and Allied Health School, Beijing Language and Culture University, Beijing, People's Republic of China; ^bInstitute of Life and Health Sciences, Beijing Language and Culture University, Beijing, People's Republic of China; ^cKey Laboratory of Language and Cognitive Science (Ministry of Education), Beijing Language and Culture University, Beijing, People's Republic of China; ^dSchool of Foreign Languages, Shanghai Jiao Tong University, Shanghai, People's Republic of China

ABSTRACT

Chunking is a key mechanism by which working memory (WM) compresses sequential information, yet little is known about how it reorganises positional representations. We examined when chunking produces stable and shared local–global positional structure in WM. Across four behavioural experiments, participants encoded nine-item sequences and later reported the original position of a probed item. The sequence types differed in the source and strength of chunk-defining boundaries: structured linguistic sequences, scrambled linguistic sequences, color-segmented shape sequences, and uniformly colored shape sequences. We characterised positional representations by applying parameterised behavioural representational similarity analysis to position-specific response distributions. Structured linguistic sequences produced a near-orthogonal geometry, indicating clearly separable local and global positional structure. Color-segmented shape sequences yielded a reduced-angle, reduced-scale geometry, suggesting weaker local–global organization; individual-level analyses further indicated that this intermediate group-level solution partly reflected participant-level heterogeneity. In contrast, scrambled linguistic and uniformly colored shape sequences were best described by one-dimensional solutions, implying that local and global dimensions were no longer separable when explicit boundary cues were absent. These findings suggest that chunking does not produce a uniform representational format in WM; instead, separable local–global structure depends on the source and strength of chunking cues.

ARTICLE HISTORY

Received 26 March 2026
Accepted 11 July 2026

KEYWORDS

Working memory; sequential memory; positional representation; hierarchical structure

Serial recall of long sequences is inherently challenging due to the capacity limits of working memory (WM). One important way to mitigate these limits is to reorganise sequential input into chunks (Nassar et al., 2018; Thalmann et al., 2019) and this strategy compresses the input sequences and leads to better retention as well as recognition (Brady et al., 2009; Chekaf et al., 2016; Musfeld et al., 2024; Planton et al., 2021).

Such chunking can arise from different sources and has been observed with many kinds of stimuli (Ellis, 2003; Fan et al., 2025; Huang & Awh, 2018; Paparella & Papeo, 2022). When reading sequentially presented words, prior linguistic knowledge helps comprehenders define chunk boundaries (Ding et al., 2016; Miyagawa et al., 2013) and the need to rapidly derive compositional meaning may encourage tighter binding of linguistic units within a chunk (e.g., within a word or multi-word expression) than across chunk boundaries. For instance, when listening to the syllable sequence “head-man-fix-up-play-ground”, listeners

are more likely to organise the input as “head-man | fix-up | play-ground” rather than as “head-man-fix | up-play-ground” to comprehend the sequence. Similarly, for non-linguistic visual materials, perceptual grouping among shapes or objects can help define chunk boundaries (Jiang et al., 2000; Parmentier et al., 2006; Woodman et al., 2003). That is, objects or shapes that are grouped together tend to be stored together within a chunk.

Even in the absence of explicit segmentation cues, participants may impose subjective organization on sequential material (R. Henson, 2001; Hitch et al., 1996). However, such self-generated grouping is likely to be weaker and less consistent across individuals than grouping supported by external boundaries, such as semantic structure in language or perceptual grouping cues in visual sequences. For instance, unfamiliar digit strings are often recoded into smaller subgroups, although the size and placement of these subjectively imposed groups may vary across individuals.

Although previous research has extensively documented the behavioural benefits of chunking (Guitard et al., 2021; Mathy et al., 2024), much less is known about the representational format through which chunking supports sequential memory. This issue is especially important for positional coding in working memory: when a sequence is chunked, are items still represented along a single linear order, or is positional information reorganised into multiple levels? Theories and computational models of serial order memory have long suggested that grouped sequences may rely on both within-group and between-group positional information (R. N. A. Henson, 1998; Hurlstone, 2019). Such accounts imply that chunking should reshape the similarity structure of sequence representations: adjacent items within the same chunk may be encoded differently from equally adjacent items that span a chunk boundary, whereas items occupying corresponding local positions in different chunks may retain partial similarity despite being globally separated.

In the present study, “global” position refers to the position of a chunk within the overall sequence, whereas “local” position refers to the position of an item within a chunk. For example, in a nine-item sequence organised as three triplets, positions 1, 4, and 7 occupy different global positions but share the same local position, because each is the first item within its triplet. Similarly, positions 2, 5, and 8 share the second local position, and positions 3, 6, and 9 share the third local position. A separable local–global representation would therefore predict that response patterns are not determined only by linear serial distance. For instance, positions 1 and 4 would be expected to show partially similar response patterns because they share the same local position, even though they occur in different chunks. Conversely, positions 3 and 4 are adjacent in the overall sequence but fall on opposite sides of a chunk boundary, and may therefore be represented as more distinct than expected from serial adjacency alone.

Recent neural work on structured linguistic sequences provides a concrete, parameterised formalization of this possibility, suggesting that chunked sequences may be encoded within a two-dimensional representational geometry, with separable codes for global position across chunks and local position within chunks (Fan et al., 2025). Under such an account, items occupying corresponding positions in different chunks may retain partial similarity that is not reducible to linear adjacency alone. This framework therefore offers a principled way to test whether chunking reorganises sequential representations into separable positional dimensions. Yet if chunking can arise from overlearned linguistic structure, from explicit perceptual grouping, or from weaker self-generated organization, then its representational consequences may not be uniform. The critical question, therefore, is under what conditions chunking gives rise to stable and shared local–global positional representations in working memory. In particular, it remains unknown whether such

separability requires rich linguistic structure, whether it can also be supported by simpler perceptual boundary cues, and whether, in the absence of explicit boundaries, sequence representations revert to more linear or more heterogeneous forms.

To address this question, we conducted four behavioural experiments and used a parameterised behavioural representational similarity analysis to infer the latent geometry of positional representations from response patterns. The logic is that positions that are represented more similarly in working memory should give rise to more similar patterns of responses. For each target position, we treated the distribution of responses across output positions as a vector in a nine-dimensional response space, with each dimension corresponding to one possible reported position, and quantified the pairwise distances between these vectors. Smaller distances indicated more similar behavioural confusion patterns, allowing us to construct a behavioral representational dissimilarity matrix and, from it, an estimated behavioural geometry of sequence positions. This approach makes it possible to test whether positional representations are better characterised by a single linear organization or by separable local and global structure.

We applied this behavioural-geometry approach to four sequence types that differed in the source and strength of chunk-defining boundaries: structured linguistic sequences, scrambled linguistic sequences, color-segmented shape sequences, and uniformly colored shape sequences. We use the term “chunk” to refer to a subset of sequence items that is separated from other items by an explicit or learned boundary cue and may therefore be encoded as a relatively coherent unit in working memory. Importantly, this definition does not imply that all chunks involve equally strong within-chunk binding. The structured linguistic condition involved overlearned lexical chunks, in which the elements within a trisyllabic word are strongly bound by lexical-semantic knowledge. By contrast, the color-segmented shape condition involved perceptual grouping, in which color provided an explicit boundary cue for segmenting the sequence but did not specify the internal order of shapes within each segment. The remaining two conditions, scrambled linguistic sequences and uniformly colored shape sequences, lacked explicit boundary cues and served as no-cue comparison conditions. This design allowed us to compare local–global positional organization supported by overlearned linguistic structure, perceptual grouping, and the absence of explicit boundary information. The procedure is illustrated in Figure 1.

We predicted that structured linguistic sequences would show the clearest evidence of stable local–global separability, because their chunk boundaries are strongly constrained by prior linguistic knowledge and support tightly integrated, highly shared chunk representations. We further predicted that color-segmented shape sequences would show weaker or less stable evidence of

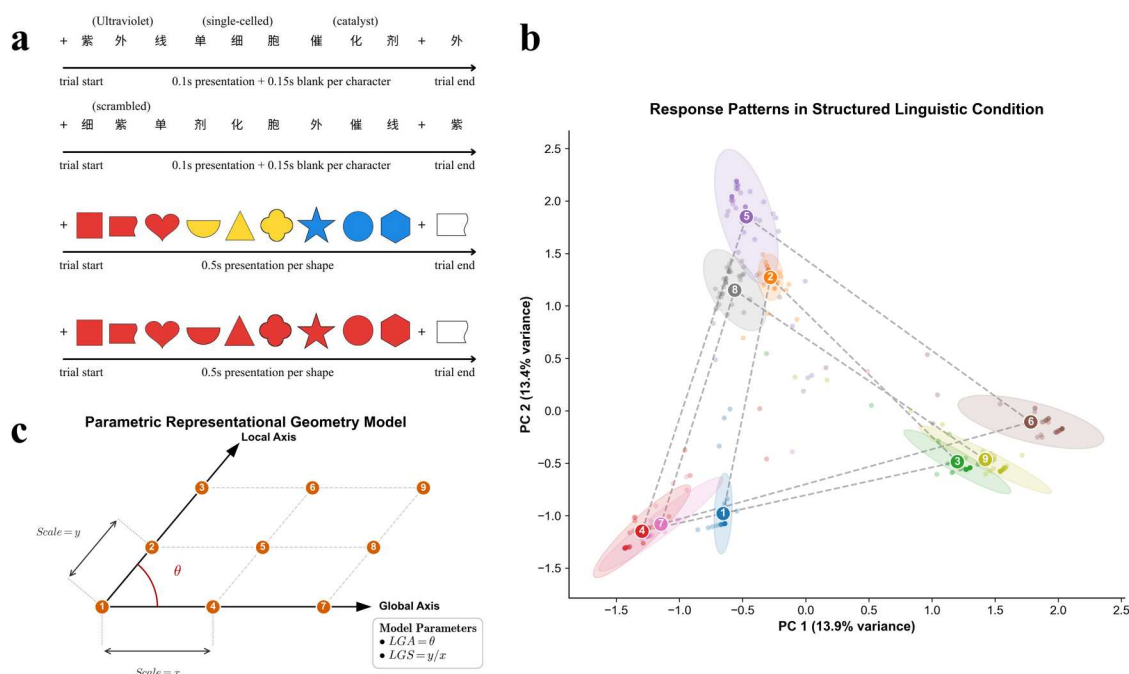


Figure 1. Illustration of the experimental procedure and analysis framework. (a) The four experimental conditions used in the study: structured linguistic materials, scrambled linguistic materials, color-segmented shapes, and uniformly colored shapes. (b) Schematic visualization of the behavioural response-vector approach. For each target position, the distribution of responses across output positions was represented as a vector in a high-dimensional response space. For visualization, the high-dimensional response data in structured linguistic condition are projected into 2D space using PCA. Each number represents the centroid of the confidence ellipse for the corresponding target position (1–9). (c) The parametric representational geometry model used to quantify the structure, defined by two free parameters: Local–Global Angle (LGA) and Local–Global Scale (LGS).

such separability. Although explicit perceptual boundaries can support grouping even in the absence of linguistic structure, they provide a shallower and potentially less constraining basis for chunk formation than overlearned semantic structure. Accordingly, we expected local and global positional information to be less distinctly separated in this condition than in the structured linguistic condition. By contrast, when explicit boundary information was unavailable, as in the scrambled linguistic and uniformly colored shape conditions, we expected sequence representations to be better characterised by more linear or more heterogeneous forms of organization.

Methods

We conducted four online behavioural experiments that differed in the source and strength of chunk-defining boundaries: a structured linguistic experiment, a scrambled linguistic experiment, a color-segmented shape experiment, and a uniformly colored shape experiment. Because the same analytic framework was applied across all four experiments, the data-analysis procedures are reported in a separate General Analysis section following the experiment-specific methods.

Ethical approval statement

This study has been reviewed and approved by the Ethics Committee of Beijing Language and Culture University

(Approval No.: 2024BYLL55). All methods were performed in accordance with the relevant guidelines and regulations. Informed consent was obtained from all participants prior to their participation in the study.

Experiment 1: structured linguistic

Participants

Forty participants were recruited for the structured linguistic experiment (age range: 18–27 years; 17 males; native Chinese speakers). To ensure high data quality given the online nature of the experiment, strict inclusion criteria were applied. Participants were recruited via campus posters and were restricted to currently enrolled university students aged 18–35 for all four experiments. The four experiments used independent, non-overlapping samples: each participant took part in only one of the four experiments.

Materials

Stimuli were adapted from Fan et al. (2025). In the linguistic experiments, each sequence consisted of nine visually presented Chinese characters. For the present materials, each character corresponded to one syllable and typically represented a morphemic unit. In the structured linguistic condition, every three consecutive characters formed a familiar trisyllabic Chinese word, such that each nine-character sequence comprised three lexical units and carried a clear lexical-semantic structure. In the scrambled linguistic

condition, the same characters were pseudo-randomly reordered to disrupt the original trisyllabic word units and eliminate the corresponding lexical-semantic hierarchy. Unlike the original auditory presentation in Fan et al. (2025), here we presented the characters visually, one at a time. Specifically, the stimulus pool contained 81 unique trisyllabic Chinese words. Across the experiment, each word was presented on average 8 times. All 81 words served as target-containing words at least once, with each word being probed on average 2.7 times.

Procedure

The four experiments were programmed using jsPsych (de Leeuw, 2015) and hosted online via GitHub Pages. The entire procedure for each experiment lasted approximately 40–60 min.

Each serial position (1–9) served as the target 24 times, yielding a total of 216 trials per participant. Each trial began with a central fixation cross. Subsequently, nine Chinese characters were presented sequentially. Each character appeared for 100 ms, followed by a 150 ms inter-stimulus interval (ISI). Following the sequence presentation, a 1000 ms retention interval ensued. Participants then performed an intervening distractor task, in which they judged whether a random number between 11 and 20 was odd or even (i.e., a parity judgment task). A second 1000 ms retention interval followed the distractor task. Finally, a target character was displayed for 500 ms, and participants identified its original serial position in the sequence by pressing the corresponding number key (1–9) on their keyboard.

Experiment 2: scrambled linguistic

Participants

Forty participants were recruited for the scrambled linguistic experiment (age range: 18–29 years; 4 males; native Chinese speakers).

Materials

The stimuli were derived from the structured linguistic sequences. Specifically, the nine characters from each structured sequence were reorganised to eliminate the inherent semantic hierarchy. The presentation order was pseudo-randomised with the constraint that no two adjacent characters from the original sequence could appear consecutively in the scrambled sequence, thereby preventing the preservation of local word boundaries.

Procedure

Each serial position (1–9) served as the target 24 times, yielding a total of 216 trials per participant. Each trial began with a central fixation cross, followed by the sequential presentation of nine Chinese characters. Each character appeared for 100 ms, followed by a 150 ms ISI. After sequence presentation, a 2000ms retention interval ensued. Unlike Experiment 1, the parity judgment task

was omitted to balance task difficulty. After the retention interval, a target character was displayed for 500 ms, and participants reported its original serial position by pressing the corresponding number key (1–9) on their keyboard.

Experiment 3: color-segmented shape

Participants

Forty participants were recruited for the color-segmented shape experiment (age range: 18–29 years; 11 males; native Chinese speakers).

Materials

The stimulus pool contained nine distinct shapes (circle, flag, heart, hexagon, quatrefoil, semicircle, square, star, and triangle) rendered in three colors (red, yellow, and blue). In this experiment, each sequence consisted of three successive red shapes, three successive yellow shapes, and three successive blue shapes. The order of the color blocks was fixed, whereas the order of shapes within each block was randomised.

Procedure

Each serial position (1–9) served as the target 24 times, yielding a total of 216 trials per participant. The procedure was broadly similar to that of the linguistic experiments, but with two modifications introduced on the basis of pilot testing to balance task difficulty. First, each shape was presented for 500 ms with no inter-stimulus interval (0 ms ISI). Second, the parity judgment task was omitted to avoid floor effects, as pilot testing indicated that the visual shape task was inherently more demanding. After sequence presentation, participants completed a 2000ms retention interval, after which a target shape was displayed for 500 ms. Participants then indicated its original serial position by pressing the corresponding number key (1–9) on their keyboard.

Experiment 4: uniformly colored shape

Participants

Forty participants were recruited for the uniformly coloured shape experiment (age range: 18–35 years; 10 males; native Chinese speakers).

Materials

The stimulus pool was identical to that used in Experiment 3 and consisted of nine distinct shapes (circle, flag, heart, hexagon, quatrefoil, semicircle, square, star, and triangle) rendered in three colors (red, yellow, and blue). In this experiment, all shapes within a given trial shared the same color, which was randomly selected from red, yellow, or blue.

Procedure

Each serial position (1–9) served as the target 24 times, yielding a total of 216 trials per participant. As in

Experiment 3, the procedure was modified on the basis of pilot testing to preserve analyzable performance. Each shape was presented for 500 ms with no inter-stimulus interval (0 ms ISI), and the parity judgment task was omitted to avoid floor effects. After sequence presentation, participants completed a 2000ms retention interval. A target shape was then displayed for 500 ms, and participants reported its original serial position by pressing the corresponding number key (1–9) on their keyboard.

Across the two shape experiments, these modifications were introduced because pilot testing indicated that the nonlinguistic shape sequences were substantially more difficult than the linguistic sequences under the original timing parameters. Using identical presentation timing and distractor demands across all experiments would therefore have resulted in near-floor performance in the shape tasks. We adjusted timing and distractor demands to preserve sufficiently informative response distributions for the representational analyses. However, because the experiments were not fully matched in temporal and interference demands, cross-experiment differences should be interpreted with appropriate caution.

General analysis

Construction of behavioural representational dissimilarity matrices (RDM)

For each participant, a 9×9 behavioural response matrix was generated by cross-tabulating the probe position against the reported position. In this matrix, the row index i corresponds to the actual target position, and the column index j corresponds to the reported position; thus, the cell (i, j) represents the frequency count of reporting position j given target i . Behavioural accuracy was calculated for each subject as the ratio of correct responses (the sum of the diagonal elements) to the total number of trials.

Following the generation of the response matrices, we computed a behavioural RDM for each participant to quantify the distinctiveness of positional representations. Specifically, the response distribution for each of the nine probe positions (i.e., each row of the 9×9 response matrix) was treated as a nine-element vector, with each element giving the frequency of reporting one of the nine possible output positions. Each target position can thus be regarded as a point in a nine-dimensional response space whose axes correspond to the nine reported positions. For example, if a participant reported positions 1, 2, and 3 with frequencies of 20, 3, and 1 when the target appeared at position 1, and never reported the remaining positions, the response vector for target position 1 would be [20, 3, 1, 0, 0, 0, 0, 0, 0]. We then calculated the pairwise Euclidean distances between these position-specific vectors, using Euclidean distance as a simple measure of overall dissimilarity in response profiles that preserves differences in both the distribution and magnitude of responses across positions.

This process yielded a symmetric 9×9 RDM for each subject, where the value in cell (i, j) represents the Euclidean distance between the response patterns for target position i and target position j . A smaller distance indicates that the two positions exhibited highly similar response profiles (implying high confusability between them), while a larger distance indicates distinct behavioural profiles.

RSA based on parametric representational geometry model

To reconstruct the latent geometric structure of the behavioural representations, we employed a parametric representational geometry model defined by two free parameters from Fan et al. (2025): the local–global angle (LGA) and the local–global scale (LGS). The LGA represents the angle between the axes encoding local and global ranks, and thus characterises the geometric relation between within-group and between-group positional information. Values near 0° or 180° indicate a largely collinear, one-dimensional organization, whereas values near 90° indicate a more nearly orthogonal and therefore more separable local–global structure. The LGS denotes the ratio of the representational distance unit along the local axis relative to the global axis. Values greater than 1 indicate that a one-unit difference along the local axis corresponds to a larger representational distance than a one-unit difference along the global axis, whereas values below 1 indicate the opposite.

We performed a grid search to identify the geometric parameters that best captured the behavioural data. A comprehensive set of candidate model RDMs was generated by systematically sampling the parameter space: the LGA was varied from 0° to 180° in increments of 1° , and the LGS was varied from 0 to 2 in increments of 0.01. For each parameter combination, the coordinates of the nine sequence positions were determined, and the pairwise Euclidean distances between them were calculated to construct the corresponding model RDM.

To estimate the latent representational geometry, we regressed the behavioural RDM against each candidate model RDM using a linear mixed-effects model (LMM) in Python-statsmodels (Seabold & Perktold, 2010). In this model, the behavioural RDM served as the dependent variable, the candidate model RDM as the fixed effect, and subjects as a random effect (random intercepts) to account for individual variability. The goodness of fit for each parameter combination was quantified using the negative log-likelihood (NLL). The parameter pair (LGA, LGS) yielding the lowest NLL was identified as the best-fitting representational geometry. Finally, we applied the Benjamini-Hochberg procedure to control the False Discovery Rate (FDR) at $\alpha = 0.05$ for the p values associated with the fixed effect of the model RDM.

To estimate the uncertainty of the geometric parameters (LGA and LGS) and derive their 95% CIs, we performed a nonparametric bootstrapping analysis with

5,000 iterations. In each iteration, participants were resampled with replacement from the original dataset to create a bootstrapped sample with the same number of participants as the original condition. For each bootstrap sample, we repeated the full grid-search procedure using the same linear mixed-effects likelihood criterion as in the main analysis. The parameter combination yielding the lowest negative log-likelihood was recorded as the best-fitting LGA and LGS for that bootstrap sample. For computational efficiency, the random-intercept linear mixed-effects likelihood was evaluated using an analytically equivalent vectorised implementation rather than repeatedly calling the full mixed-model fitting routine for every candidate model and bootstrap sample. Importantly, both the point estimates and the bootstrap confidence intervals were therefore obtained using the same LMM-based negative log-likelihood criterion. This procedure resulted in empirical sampling distributions for both parameters. The 95% CIs were then determined using the percentile method, defined as the range between the 2.5th and 97.5th percentiles of the bootstrapped distributions.

Individual-level RSA

To assess whether the group-level geometries reflected relatively consistent participant-level representational structures or instead arose from a mixture of heterogeneous participant-level patterns, we conducted an exploratory individual-level RSA using the same parametric geometry model. For each participant, we fitted the model to that participant's behavioural RDM by grid-searching over LGA from 0° to 180° in 1° steps and LGS from 0 to 2 in 0.01 steps. The best-fitting parameter combination was defined as the one that maximised the Pearson correlation between the participant-level behavioural RDM and the candidate model RDM. Because the linear mixed-effects framework used in the main group-level analysis was designed to estimate condition-level effects while treating participants as a random factor, it was not directly applicable to single-participant fits. We therefore used correlation-based fitting for this exploratory individual-level analysis. We then examined the distribution of the resulting per-participant LGA and LGS estimates within each of the four conditions. Because LGA values of 0° and 180° both correspond to collinear, one-dimensional solutions, we additionally transformed each LGA estimate into a separability angle, defined as $\min(\text{LGA}, 180^\circ - \text{LGA})$. This folded measure ranged from 0° to 90° , with 0° indicating a one-dimensional geometry and 90° indicating an orthogonal, maximally separable local-global geometry. Because participant-level RDMs were estimated from a limited number of trials and are therefore considerably noisier than the group-level data, these analyses were treated as exploratory.

In addition to the analyses described above, two supplementary analyses are reported in the Supplementary Materials. First, we conducted a parameter stability analysis to evaluate the robustness of the estimated

representational geometry and the adequacy of the sample size. Second, we conducted a sensitivity analysis in which the accuracy of the structured linguistic condition was reduced by amplifying each participant's own error patterns, after which the RSA was refitted.

Results

Descriptive structure of behavioural response patterns

The structured linguistic condition exhibited the highest accuracy (0.87 ± 0.17 , mean $\pm$ SD), followed by the color-segmented shape condition (0.48 ± 0.21), scrambled linguistic condition (0.48 ± 0.17) and uniformly colored shape condition (0.43 ± 0.17). Given that all participants performed above chance level ($1/9$) and that the distribution of error responses was essential for reconstructing the behavioural representational geometry, no data were excluded. The response probability matrices for the four conditions are shown in Figure 2.

To provide an initial descriptive view of the response structure, we visualised the position-specific behavioural response distributions in a two-dimensional space using principal component analysis (PCA; Figure 3). This visualization was intended purely for illustration and did not enter into the formal model fitting. In the structured linguistic condition, the projected centroids suggested an organization broadly consistent with separable local and global structure, with positions occupying corresponding roles across chunks appearing relatively close in the projected space. A related but less differentiated pattern was visible in the color-segmented shape condition, where positions within the same perceptual group appeared more tightly arranged than positions spanning color-defined boundaries. By contrast, this pattern was largely absent in the scrambled linguistic and uniformly colored shape conditions, where the projected positions appeared more uniformly spaced across adjacent serial positions, without the boundary-related separation apparent in the structured sequences. These descriptive patterns motivated the subsequent representational dissimilarity and parametric geometry analyses.

RSA based on parametric representational geometry model

The group-averaged behavioural RDMs were shown in Figure 4. For all experimental conditions, the best-fitting geometric models significantly predicted behavioural response patterns (all corrected $p < 0.001$). As illustrated in Figure 4, the structured linguistic condition yielded optimal parameters of $\text{LGA} = 87^\circ$ (95% CI: $[77^\circ, 97^\circ]$) and $\text{LGS} = 1.35$ (95% CI: $[1.04, 1.77]$), indicating a near-orthogonal solution consistent with strongly separable local and global positional structure. The LGS value above 1 indicates that, in the best-fitting solution, a

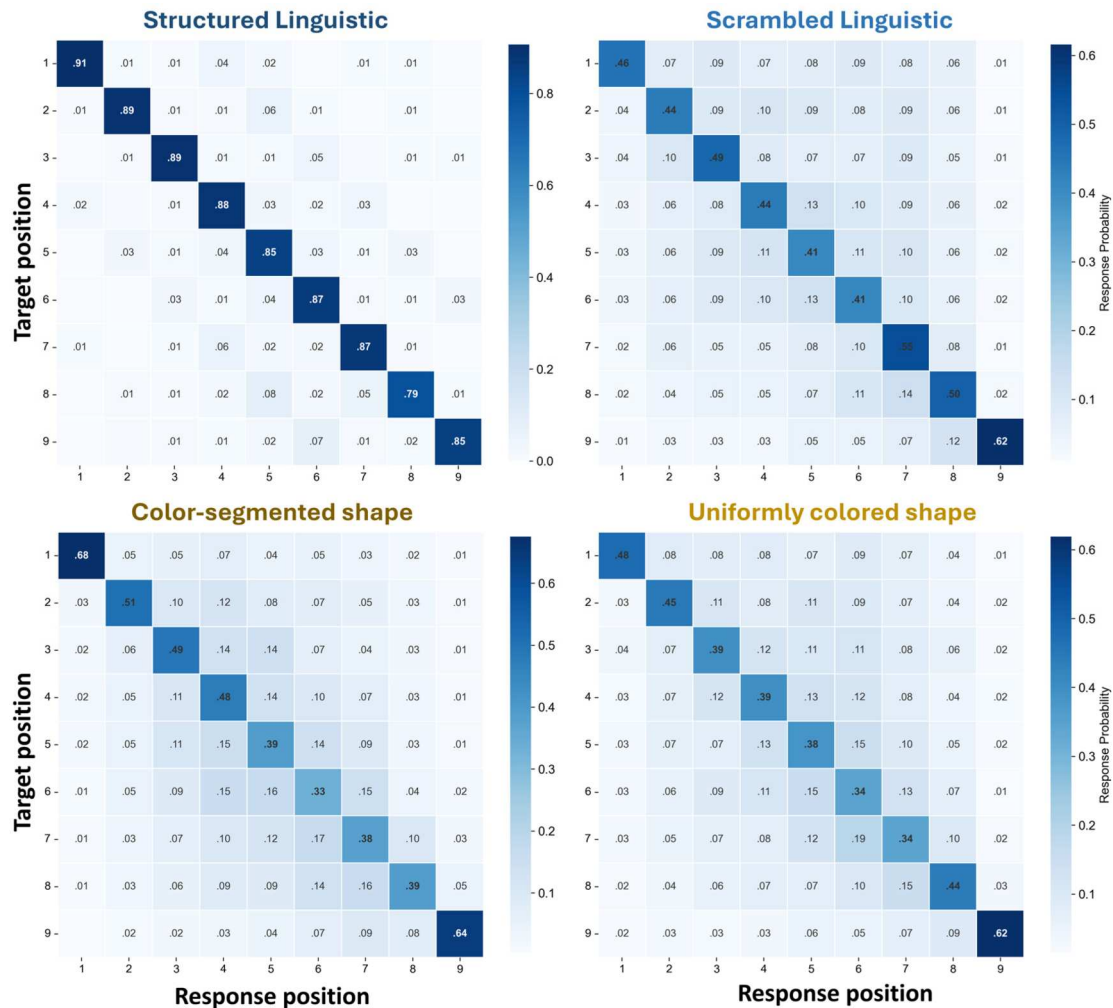


Figure 2. Response probability matrices across the four conditions. Rows indicate target positions, and columns indicate reported positions. The diagonal elements indicate positional accuracy, whereas the off-diagonal elements reflect the patterns of positional errors.

one-unit difference in local rank corresponded to a larger representational distance than a one-unit difference in global rank. The color-segmented shape condition yielded an intermediate solution, with best-fitting parameters of $LGA = 43^\circ$ (95% CI: $[19^\circ, 62^\circ]$) and $LGS = 0.57$ (95% CI: $[0.46, 0.68]$). Relative to the structured linguistic condition, this reduced-angle, reduced-scale solution is consistent with a less differentiated form of local–global organization. The LGS value below 1 indicates that local-rank differences were compressed relative to global-rank differences in the recovered geometry. In contrast, both unstructured control conditions were best captured by a robust 1-D geometry. Specifically, the uniformly colored shape condition converged to $LGA = 0^\circ$ (95% CI: $[0^\circ, 0^\circ]$) and $LGS = 0.59$ (95% CI: $[0.54, 0.64]$). Similarly, the scrambled linguistic condition yielded parameters of $LGA = 0^\circ$ (95% CI: $[0^\circ, 0^\circ]$) and $LGS = 0.60$ (95% CI: $[0.54, 0.65]$), indicating that under the present task conditions, removal of the original linguistic structure was associated with a substantially more linearised representational organization.

Notably, in both unstructured conditions, the recovered 1-D solutions ($LGA = 0^\circ$, $LGS \approx 0.6$) also implied a violation of ordinal monotonicity. Under this parameter range, representational distance did not increase monotonically with serial order; as illustrated in Figure 4, position 4 was reconstructed as being closer to positions 1 and 2 than to the immediately preceding position 3. This pattern was observed in both unstructured conditions despite their differing stimulus materials, suggesting a stable feature of the recovered geometry when explicit boundary cues were absent. We return to the theoretical implications of this non-monotonic pattern in the Discussion.

Individual-level RSA

As shown in Figure 5, individual-level estimates differed across conditions. Separable local–global geometry was most evident in the structured linguistic condition (median separability angle = 68° , $IQR = 48^\circ\text{--}77^\circ$; $50\% \geq 70^\circ$), although considerable individual variation was still observed. Both unstructured conditions were

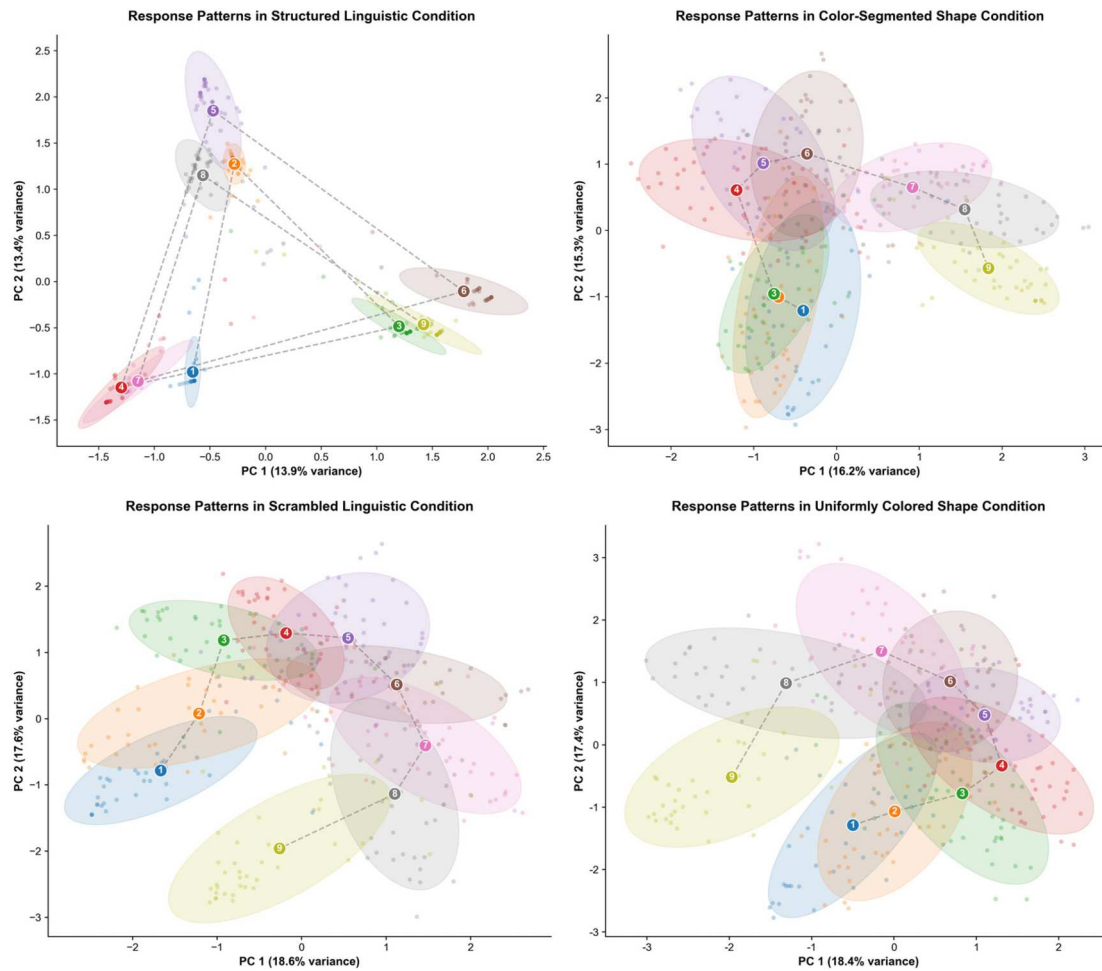


Figure 3. Descriptive visualization of participants' behavioural response distributions across the four conditions. High-dimensional response data were projected into 2D space using PCA for visualization only and were not used in the subsequent model fitting. Each number represents the centroid of the confidence ellipse for the corresponding target position (1–9). Each small point represents the response distribution of a single participant for a given target position; each condition therefore contains $9 \times 40 = 360$ points.

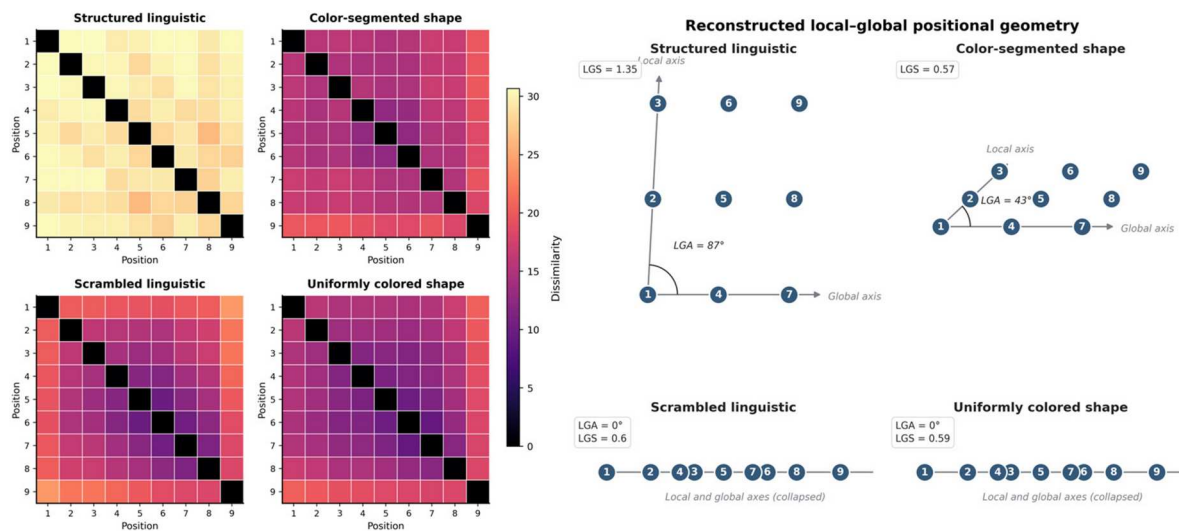


Figure 4. Group-averaged behavioural RDMs across the four conditions (left) and geometric reconstructions (right). Each matrix in the left panel shows the average pairwise Euclidean distances between position-specific behavioural response distributions across participants. The right panel illustrates the reconstructed representational geometry corresponding to the optimal parameters.

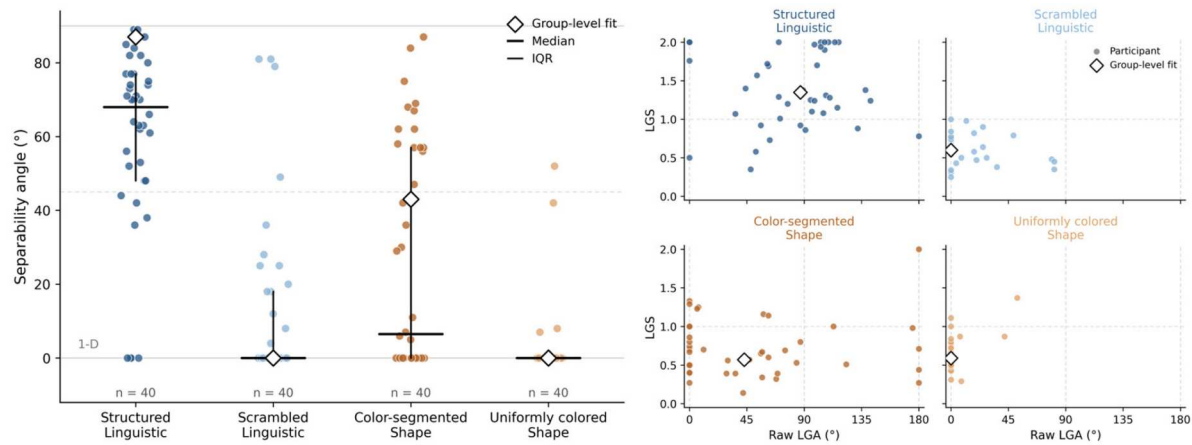


Figure 5. Individual-level RSA estimates. Left: separability angle derived from individual-level LGA estimates. A value of 0° indicates a one-dimensional solution, whereas 90° indicates an orthogonal, maximally separable local–global geometry. Right: scatter plots of individual-level LGA and LGS estimates.

predominantly one-dimensional (scrambled linguistic: median = 0° , IQR = 0° – 18° ; uniformly colored shape: median = 0° , IQR = 0° – 0°). Notably, although the color-segmented shape condition yielded an intermediate group-level estimate (LGA = 43°), its median participant-level separability angle was only 6.5° (IQR = 0° – 57°), indicating that many participants were close to a one-dimensional solution, whereas a subset showed more separable structure. LGS was likewise largest in the structured linguistic condition (median = 1.30) and smaller in the other conditions (medians = 0.56–0.67). Because participant-level RDMs were relatively noisy, these individual-level results should be interpreted as exploratory and descriptive.

Discussion

The present study asked under what conditions chunking gives rise to stable and shared local–global positional representations in working memory. Across four sequence types, the results suggest that such separability was strongest when chunk boundaries were strongly constrained by linguistic structure, weaker when grouping was supported only by perceptual cues, and largely absent when explicit boundary information was unavailable. Exploratory individual-level analyses broadly supported this graded pattern, while also showing that the intermediate group-level estimate in the color-segmented shape condition may have arisen largely from participant-level heterogeneity. Thus, the findings are more consistent with a graded account of boundary-constrained sequence organization than with a simple distinction between “hierarchical” and “non-hierarchical” representations. Under the present task conditions, local–global positional structure appeared most clearly when chunk boundaries were strongly specified and most linearly organised when they were not.

A key question concerns the difference between the two conditions that contained explicit boundary cues. The structured linguistic condition yielded a near-

orthogonal solution (LGA = 87°), whereas the color-segmented shape condition yielded a substantially reduced-angle solution (LGA = 43°). We interpret this contrast as reflecting a difference in the strength and consistency with which local and global positional information became separable, rather than as evidence for two qualitatively different forms of 2-D organization. Linguistic structure provides overlearned, meaning-bearing, and highly shared constraints on chunk formation: the characters within a word are bound by lexical-semantic structure that strongly constrains how the sequence is parsed. By contrast, color boundaries in the shape condition provided an explicit grouping cue, but a shallower and less constraining basis for chunk formation. The individual-level RSA further supported this interpretation: although the color-segmented shape condition produced an intermediate group-level LGA, many participants showed separability estimates closer to a one-dimensional solution, whereas a subset showed more separable structure. Thus, the reduced-angle group-level solution in this condition may reflect participant-level heterogeneity in the use of perceptual grouping cues, rather than a uniformly intermediate local–global code shared by all participants.

The contrast between the two conditions may also reflect the different information provided by lexical structure and color segmentation. In the structured linguistic condition, each chunk corresponded to a familiar trisyllabic word. Because the syllables within a word are strongly bound by learned lexical-semantic structure, a probed syllable provides information not only about the chunk to which it belongs but also about its position within that word. This may reduce within-chunk positional confusions and support a more clearly separable local–global geometry. In contrast, color segmentation provides a strong cue to chunk membership but does not specify the order of items within a chunk. For example, a red shape may be confidently assigned to the red segment, while its exact position within that segment remains more confusable. This difference may explain why the color-segmented

shape condition showed a reduced-angle and reduced-scale solution rather than the near-orthogonal structure observed for linguistic sequences.

The LGS estimates further suggest that the two conditions differed not only in the separability of local and global dimensions, but also in their relative weighting. In the structured linguistic condition, $LGS > 1$ indicates that a one-unit difference in local rank corresponded to a larger representational distance than a one-unit difference in global rank. This pattern is consistent with the idea that positions within a linguistic chunk were more sharply differentiated, even as the chunk itself provided a stable and highly shared unit of organization. By contrast, in the color-segmented shape condition, $LGS < 1$ indicates that local-rank differences were compressed relative to global-rank differences. This is consistent with the idea that color cues were sufficient to define perceptual groups, but less effective in supporting fine-grained differentiation among positions within those groups.

If the contrast between the two structured conditions speaks to differences in the strength of boundary-constrained organization, the two unstructured conditions speak to what happens when explicit boundary support is absent. Despite differences in stimulus type and timing parameters, both the scrambled linguistic and uniformly colored shape conditions converged to $LGA = 0^\circ$, indicating that local and global positional information no longer formed separable dimensions under the present task conditions. Importantly, however, this should not be interpreted as evidence for a complete absence of local-position-related information. When $LGA = 0^\circ$, the local and global axes are collinear rather than separable; therefore, residual local-rank-related structure can still be expressed within a one-dimensional solution through the LGS parameter. This explains why the first item of a nominal triplet, especially positions 4 and 7, can appear left-shifted in the recovered geometry. In this sense, the absence of explicit boundaries did not simply eliminate local-position-related structure, but instead prevented local and global positional information from forming independent dimensions. We do not interpret this non-monotonic pattern as evidence that participants literally reversed the sequence order. Rather, it suggests that in the absence of explicit boundary cues, positional organization may become weakly constrained across trials and participants. Under such conditions, the group-level best-fitting geometry may no longer reflect a single stable serial chain, but instead a distorted 1-D solution in which residual local-rank-related variation is expressed along the same dimension as global order.

Taken together, the present findings help clarify the conditions under which chunking gives rise to stable and shared local–global positional representations in working memory. Local–global separability was strongest when chunk boundaries were strongly constrained by over-learned linguistic structure, weaker when boundaries were supported only by perceptual grouping cues, and

absent or distorted when explicit boundary information was unavailable. These results therefore suggest that chunking does not uniformly produce the same representational format. Rather, the structure of positional coding appears to depend on the source and strength of the constraints that define chunk boundaries.

Limitations

Several limitations should be noted. First, the four conditions differed not only in the availability of chunk-defining boundaries but also in temporal parameters and the presence or absence of a distractor task, because these adjustments were introduced to avoid floor effects and preserve analyzable response distributions. We therefore cannot attribute all cross-condition differences exclusively to boundary structure. At the same time, the close convergence of the two no-cue conditions suggests that the absence of explicit boundary support was likely related to the observed linearised organization. In addition, the linguistic and shape conditions differed in their item-reuse structure. Individual lexical items were reused less frequently than individual shape identities, because the shape experiments relied on a pool of only nine shape identities that recurred throughout the experiment. This difference may have contributed to the observed cross-domain differences between the linguistic and shape conditions. However, it is unlikely to account for the overall pattern of results, because the structured and scrambled linguistic conditions shared the same character pool, and the color-segmented and uniformly colored shape conditions shared the same shape pool. The within-domain contrasts therefore still point to a role of boundary structure, although future studies should more closely match item-reuse rates across stimulus domains.

Second, performance differed substantially across conditions, with the structured linguistic condition yielding fewer errors than the other conditions. Because the present analyses were based on response distributions, differences in accuracy may affect geometry reconstruction in multiple ways. Lower-accuracy conditions may include more random or heterogeneous errors, whereas the high-accuracy structured linguistic condition provides fewer off-diagonal responses from which to estimate the structure of positional confusions. Thus, both low and high accuracy can complicate the interpretation of recovered representational geometries. Because the other three conditions had broadly comparable accuracy levels, ranging from approximately 0.43–0.48, we conducted an exploratory sensitivity analysis to examine whether the recovered geometry in the structured linguistic condition remained stable when overall accuracy was reduced to levels comparable to the lower-accuracy conditions. Starting from the original response matrices in this condition (mean accuracy = 0.87), we reduced the diagonal response probabilities for each participant and redistributed the displaced probability mass across the off-diagonal cells in

proportion to that participant's own error pattern, using the position-specific group-average error pattern for the few rows without errors. This procedure preserved where errors tended to occur while increasing the overall error rate. Across degraded accuracy levels from approximately 0.80–0.40, including levels comparable to or lower than those observed in the lower-accuracy conditions, the best-fitting LGA remained close to orthogonal (87° – 91°), and LGS remained above 1. For details, see the Supplementary Materials. Because this manipulation preserves the structure of errors by design, it should not be interpreted as a fully accuracy-matched control. Nevertheless, these observations provide some reassurance that the structured linguistic geometry reflected systematic aspects of the behavioural response distributions, although future work should test this issue more directly under more closely matched performance levels.

Third, the four experiments in the current study were conducted sequentially rather than as a single randomised multi-condition study; consequently, participants were not randomly assigned to conditions. Because the conditions were not fully matched in procedure or accuracy, cross-condition differences should be interpreted descriptively rather than as causal effects of sequence condition in a fully randomised experimental sense. Future work should test whether the present pattern replicates under randomised between-subjects or mixed within-subjects designs with more closely matched procedures. A further limitation is that the same constrained parametric geometry was applied to both structured and unstructured conditions. This choice ensured that the fitted LGA and LGS parameters were directly comparable across conditions, but it also imposed assumptions about the underlying representational space. Future work could therefore use more flexible RSA models, including models with additional free parameters for position-specific distances or less constrained geometric embeddings, to test whether the observed one-dimensional solutions remain robust under more general assumptions.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

Data and code are available at <https://osf.io/dq78m>. The experiment codes are available at <https://github.com/1angxingyu/MyExperiment>.

References

- Brady, T. F., Konkle, T., & Alvarez, G. A. (2009). Compression in visual working memory: Using statistical regularities to form more efficient memory representations. *Journal of Experimental Psychology: General*, 138(4), 487–502. <https://doi.org/10.1037/a0016797>
- Chekaf, M., Cowan, N., & Mathy, F. (2016). Chunk formation in immediate memory and how it relates to data compression. *Cognition*, 155, 96–107. <https://doi.org/10.1016/j.cognition.2016.05.024>
- de Leeuw, J. R. (2015). Jpsych: A JavaScript library for creating behavioural experiments in a web browser. *Behavior Research Methods*, 47(1), 1–12. <https://doi.org/10.3758/s13428-014-0458-y>
- Ding, N., Melloni, L., Zhang, H., Tian, X., & Poeppel, D. (2016). Cortical tracking of hierarchical linguistic structures in connected speech. *Nature Neuroscience*, 19(1), 158–164. <https://doi.org/10.1038/nn.4186>
- Ellis, N. C. (2003). Constructions, chunking, and connectionism: The emergence of second language structure. In Catherine J. Doughty & Michael H. Long (Eds.), *The handbook of second language acquisition* (pp. 63–103). John Wiley & Sons, Ltd <https://doi.org/10.1002/9780470756492.ch4>
- Fan, Y., Wang, M., Fang, F., Ding, N., & Luo, H. (2025). Two-dimensional neural geometry underpins hierarchical organization of sequence in human working memory. *Nature Human Behaviour*, 9(2), 360–375. <https://doi.org/10.1038/s41562-024-02047-8>
- Guitard, D., Saint-Aubin, J., & Cowan, N. (2021). Grouping effects in immediate reconstruction of order and the preconditions for long-term learning. *Quarterly Journal of Experimental Psychology*, 75(1), 70–97. <https://doi.org/10.1177/17470218211030825>
- Henson, R. (2001). Serial order in short-term memory. *The Psychologist*, 14(2), 70–73.
- Henson, R. N. A. (1998). Short-term memory for serial order: The start-end model. *Cognitive Psychology*, 36(2), 73–137. <https://doi.org/10.1006/cogp.1998.0685>
- Hitch, G. J., Burgess, N., Towse, J. N., & Culpin, V. (1996). Temporal grouping effects in immediate recall: A working memory analysis. *The Quarterly Journal of Experimental Psychology A: Human Experimental Psychology*, 49(1), 116–139. <https://doi.org/10.1080/027249896392829>
- Huang, L., & Awh, E. (2018). Chunking in working memory via content-free labels. *Scientific Reports*, 8(1), 23. <https://doi.org/10.1038/s41598-017-18157-5>
- Hurlstone, M. J. (2019). Functional similarities and differences between the coding of positional information in verbal and spatial short-term order memory. *Memory*, 27(2), 147–162. <https://doi.org/10.1080/09658211.2018.1495235>
- Jiang, Y., Olson, I. R., & Chun, M. M. (2000). Organization of visual short-term memory. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 26(3), 683–702. <https://doi.org/10.1037/0278-7393.26.3.683>
- Mathy, F., Friedman, O., & Gauvrit, N. (2024). Can compression take place in working memory without a central contribution of long-term memory? *Memory & Cognition*, 52(8), 1726–1736. <https://doi.org/10.3758/s13421-023-01474-8>
- Miyagawa, S., Berwick, R. C., & Okanoya, K. (2013). The emergence of hierarchical structure in human language. *Frontiers in Psychology*, 4, 71. <https://doi.org/10.3389/fpsyg.2013.00071>
- Musfeld, P., Dutli, J., Oberauer, K., & Bartsch, L. M. (2024). Grouping in working memory guides chunk formation in long-term memory: Evidence from the Hebb effect. *Cognition*, 248, 1–20. <https://doi.org/10.1016/j.cognition.2024.105795>
- Nassar, M. R., Helmers, J. C., & Frank, M. J. (2018). Chunking as a rational strategy for lossy data compression in visual working memory. *Psychological Review*, 125(4), 486–511. <https://doi.org/10.1037/rev0000101>
- Paparella, I., & Papeo, L. (2022). Chunking by social relationship in working memory. *Visual Cognition*, 30(5), 354–370. <https://doi.org/10.1080/13506285.2022.2064950>
- Parmentier, F. B. R., Andrés, P., Elford, G., & Jones, D. M. (2006). Organization of visuo-spatial serial memory: Interaction of temporal order with spatial and temporal grouping. *Psychological Research Psychologische Forschung*, 70(3), 200–217. <https://doi.org/10.1007/s00426-004-0212-7>
- Planton, S., Kerkoerle, T. van, Abbi, L., Maheu, M., Meyniel, F., Sigman, M., Wang, L., Figueira, S., Romano, S., & Dehaene, S.

- (2021). A theory of memory for binary sequences: Evidence for a mental compression algorithm in humans. *PLOS Computational Biology*, 17(1), e1008598. <https://doi.org/10.1371/journal.pcbi.1008598>
- Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with python. *scipy*, 92–96. <https://doi.org/10.25080/Majora-92bf1922-011>
- Thalmann, M., Souza, A. S., & Oberauer, K. (2019). How does chunking help working memory? *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 45(1), 37–55. <https://doi.org/10.1037/xlm0000578>
- Woodman, G. F., Vecera, S. P., & Luck, S. J. (2003). Perceptual organization influences visual working memory. *Psychonomic Bulletin & Review*, 10(1), 80–87. <https://doi.org/10.3758/BF03196470>